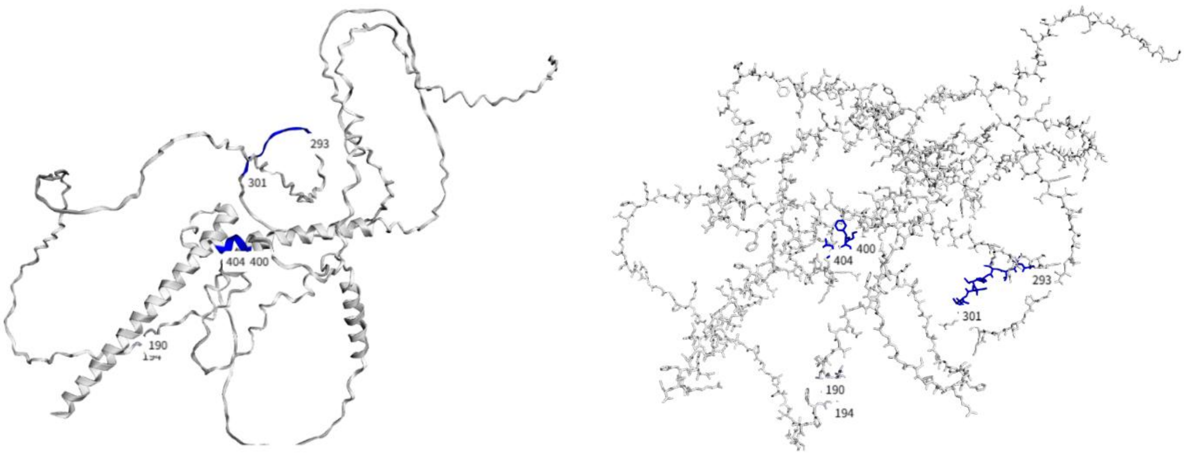
Software

The design of FuncPept is one of our core initiatives. To better facilitate FuncPept design, we have developed a software suite integrating multiple functions. This software enables the accurate prediction and visualization of Aggregation-Prone Regions (APRs) in any given protein. Concurrently, we design specific FuncPepts targeting these APRs, and subsequently validate and screen for the optimal candidates capable of inducing amyloid aggregation in the target proteins. Our core workflows are detailed below.

TANGO Prediction and Result Visualization

We primarily rely on the TANGO algorithm to predict protein APRs. TANGO evaluates and predicts APRs by scoring each amino acid within the input protein sequence. A contiguous stretch of amino acids scoring above a predefined threshold identifies an APR within the protein. Following the TANGO prediction, we map these findings onto AlphaFold-predicted structures to achieve result visualization. This process is fully automated; users simply input the protein's UniProt ID, and the software directly generates the visual output. The detailed results are shown in the figure below:



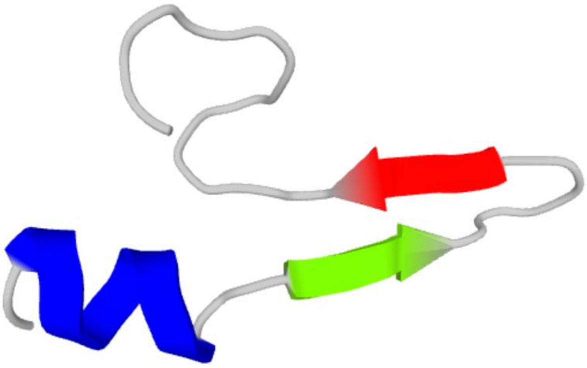
WALZE Scoring and FuncPept Screening

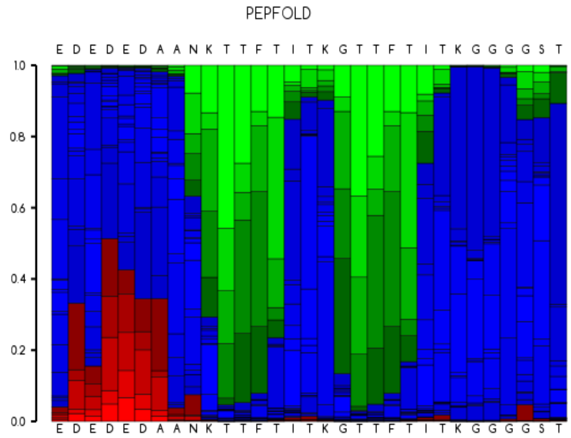
Once TANGO identifies the APR of the target protein, we apply a sliding window approach with a step size of one amino acid to locate the optimal 5-to-7 amino acid window, generating several candidate FuncPepts. These candidates are then evaluated using the WALZE scoring system. Because WALZE outputs each amino acid window individually, we filter the output based on sequence length to eliminate partial fragments and ensure only complete FuncPepts are retained. The FuncPept with the highest WALZE score is selected as our final candidate. This screening process is tightly integrated with the upstream prediction; entering the UniProt sequence in the initial step automatically triggers this workflow and generates these results.

Sequence	Average Score
TTFTITKGTTFTITK	93.31104
TTFTITGKTTFTITK	93.31104
KAVTTFTIGKTTFTITK	91.44206
TTFTITKGAVTTFTIK	90.2592
TTFTITGKAVTTFTIK	90.15468
KAVTTFTIKGTTFTITK	90.06492
TTFTITKGAVTTFTK	89.47603
TTFTITGKAVTTFTK	89.36455
KAVTTFTIGKAVTTFTIK	88.81457
TTFTITGKTTFTITVK	88.04348
TTFTITKGTTFTITVK	88.04348
KAVTTFTIGKAVTTFTK	88.03856
TTFTITKGTFTITVK	87.80379
KAVTTFTIKGAVTTFTIK	87.79264
TTFTITGKTFTITVK	87.60312
KAVTTFTIKGAVTTFTK	86.95652
KAVTTFTIGKTTFTITVK	86.86362
KAVTTFTIGKTFTITVK	86.48436
KAVTTFTIKGTTFTITVK	85.56299
KAVTTFTIKGTFTITVK	85.48102
KAVTTFTKGTFTITVK	84.44816
KAVTTFTGKTFTITVK	84.26003
TTFTITVKGTFTITVK	82.88043
KTFTITVKGTFTITVK	82.79682
TTFTITVGKTFTITVK	82.69231
KTFTITVGKTFTITVK	82.608

FuncPept Structure Prediction via PepFold

After selecting the most optimal FuncPept utilizing WALZE, we append a Tuftsin sequence (consisting of four amino acids) to the FuncPept via a Glycine-Serine (GGGGS) linker. The complete sequence is then submitted to PepFold for structural prediction. The output provides the structure of the FuncPept along with a structural propensity profile for each amino acid, serving as a reverse validation to confirm the rationale of our design. Through the integration of these predictive tools, we successfully obtain a well-designed peptide with a complete structural profile. Notably, this entire pipeline operates completely automatically.





System Compatibility

To provide a better user experience, we have adapted our software for both Linux and Windows platforms.